

Housing Reallocation and Demographic Transitions

A Two-Generation OLG Model

1 Analytical Model

This section isolates two implications of housing access in a two-period overlapping-generations economy. First, a financing-constrained young buyer can value family-sized space more than an old incumbent who would realize a taxable gain by selling it. Second, a permanent change in desired fertility can move the economy between positive steady states, so house-price appreciation and taxable gains can arise along the transition even though they vanish at both endpoints. Appendix A provides the household solutions, conditional transition results, and a high-accuracy mixed-tenure transition approximation.

1.1 Environment and household choices

At date t , the economy contains Y_t young households and O_t old households. A young household has income y_i , liquid wealth b_i , and chooses consumption c , housing h , completed fertility n , saving a' , and tenure $m \in \{R, O\}$. The pair (y_i, b_i) is drawn from a fixed distribution F in each cohort; estates are warm-glow expenditures rather than the source of the next cohort's b_i . A child requires $\chi > 0$ units of goods and $\kappa > 0$ units of space. Young and old flow utilities are

$$u_t^Y(c, h, n) = \log c + \alpha \log(h - \kappa n) + \vartheta_t \log n, \quad (1)$$

$$u^O(c^O, h^O, e) = \log c^O + \gamma \log h^O + \omega_B \log e, \quad (2)$$

where e is a warm-glow estate. A child born at t produces $\nu > 0$ young households at $t + 1$, after survival and household formation. Thus $Y_{t+1} = \nu \bar{n}_t Y_t$ and $O_{t+1} = Y_t$, where $\bar{n}_t$ is average fertility among the young.

The fixed housing stock is $\bar{H}$. Rental units satisfy $h \leq \bar{h}^R$, owner units satisfy $h \leq \bar{h}^O$, and $\bar{h}^R < \bar{h}^O$. A one-period bond has price $q \in (0, 1)$ and gross return $R = q^{-1}$. Competitive housing intermediation implies the rental user cost

$$r_t = (1 + \tau^p)P_t - qP_{t+1}, \quad \ell_t = \tau^g[P_t - P_{t-1}]_+, \quad (3)$$

where P_t is the house price, τ^p the property-tax rate, and ℓ_t the capital-gains tax per unit sold by an old owner. Housing retained until death receives a stepped-up basis; intermediaries mark their basis to market. A young buyer finances the share $\phi \in (0, 1)$ and therefore posts $(1 - \phi)P_t h$ at purchase.

Conditional on renting, the young household pays $r_t h$ and enters old age with financial wealth Ra' . Conditional on owning, it solves

$$W_t^O(i) = \max_{c, h, n, a'} u_t^Y(c, h, n) + \beta V_{t+1}^O(a_{t+1}, h) \quad (4)$$

$$\begin{aligned} \text{s.t. } c + \chi n + a' + [(1 - \phi) + \tau^p]P_t h &= y_i + b_i + T_t, \\ a_{t+1} &= Ra' - R\phi P_t h, \quad (1 - \phi)P_t h \leq b_i, \quad h \leq \bar{h}^O, \quad a' \geq 0, \end{aligned}$$

and the renter problem $W_t^R(i)$ replaces the two owner cash-flow equations with $c + \chi n + a' + r_t h = y_i + b_i + T_t$, $h \leq \bar{h}^R$, and $a' \geq 0$. Only b_i is pledgeable at closing. The transfer T_t is the equal per-household rebate of housing-tax revenue.

An old renter spends $a + T_t$ on c^O , rental housing, and the estate. An old incumbent enters with financial wealth a and inherited housing H and has the flow-equivalent budget

$$c^O + qe + (r_t - \ell_t)h^O = a + (P_t - \ell_t)H + T_t, \quad 0 < h^O \leq H, \quad e \geq P_{t+1}h^O. \quad (5)$$

The marginal cost of retaining a unit is therefore $r_t - \ell_t$: selling realizes the current tax, while retaining the unit avoids it.

To keep both tenures active without altering conditional policies, households receive iid type-I extreme-value tenure tastes with scale $\kappa_T > 0$ and an ownership intercept $\bar{\xi}$. The owner share for type i is

$$\pi_t^O(i) = \frac{\exp\{[W_t^O(i) + \bar{\xi}]/\kappa_T\}}{\exp\{W_t^R(i)/\kappa_T\} + \exp\{[W_t^O(i) + \bar{\xi}]/\kappa_T\}}. \quad (6)$$

The taste shocks smooth aggregate tenure choice; they do not enter the conditional renter or owner first-order conditions.

1.2 Housing access and conditional policies

Let $A = 1 + \gamma + \omega_B$, $k = \beta A$, and define lifetime resources and tenure-specific user costs by

$$\mathcal{W}_{it} = y_i + b_i + T_t + qT_{t+1}, \quad r_t^R = r_t, \quad r_t^O = r_t + q\ell_{t+1}. \quad (7)$$

When young saving and next-period old choices are interior, the Euler equation reduces each conditional problem to

$$(1 + k)c + \chi n + r_t^m h = \mathcal{W}_{it}. \quad (8)$$

Writing $B_t = 1 + k + \alpha + \vartheta_t$, unconstrained policies are

$$c_t^m = \frac{\mathcal{W}_{it}}{B_t}, \quad n_t^m = \frac{\vartheta_t \mathcal{W}_{it}}{B_t(\chi + \kappa r_t^m)}, \quad h_t^m = \frac{\alpha \mathcal{W}_{it}}{B_t r_t^m} + \kappa n_t^m. \quad (9)$$

The effective housing limits are

$$\hat{h}_{it}^R = \bar{h}^R, \quad \hat{h}_{it}^O = \min\left\{\bar{h}^O, \frac{b_i}{(1 - \phi)P_t}\right\}. \quad (10)$$

If the unconstrained policy exceeds $\hat{h}_{it}^m$, housing equals the limit and fertility is the unique interior root

$$\frac{\vartheta_t}{n} - \frac{\alpha \kappa}{\hat{h}_{it}^m - \kappa n} - \frac{(1 + k)\chi}{\mathcal{W}_{it} - r_t^m \hat{h}_{it}^m - \chi n} = 0. \quad (11)$$

The housing first-order condition prices the access constraint in goods:

$$\frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_{it}^m, \quad \zeta_{it}^{O,F} = \frac{\mu_{it}}{\lambda_{it}}(1 - \phi)P_t, \quad (12)$$

where λ_{it} is the marginal value of current resources and μ_{it} is the multiplier on the down-payment constraint. When the owner-size limit is slack, $\zeta_{it}^{O,F}$ is the entire owner access wedge.

1.3 Positive steady states and transitions

Let $\mathcal{N}(P, T; \vartheta)$ be average fertility and let $\mathcal{H}^Y(P, T; \vartheta)$ and $\mathcal{H}^O(P, T; \vartheta)$ be housing per young and old household after tenure choice and the induced old-state distribution. Write $\mathcal{H} = \mathcal{H}^Y + \mathcal{H}^O$. Market clearing and the government budget are

$$Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}, \quad (13)$$

$$(Y_t + O_t)T_t = \tau^p P_t \bar{H} + \ell_t O_t \bar{s}_t^O, \quad (14)$$

where $\bar{s}_t^O$ is housing sold by old owners per old household, assigning zero sales to renters.

Proposition 1 (Positive steady state). *Every positive steady state is a root $(P^*, T^*) \in \mathbb{R}_{++}^2$ of*

$$\nu \mathcal{N}(P, T; \vartheta) - 1 = 0, \quad T - \frac{\tau^p P}{2} \mathcal{H}(P, T; \vartheta) = 0. \quad (15)$$

Given a root, $Y^ = O^* = \bar{H} / \mathcal{H}(P^*, T^*; \vartheta)$ and $\ell^* = 0$. Hence every positive steady state has average fertility $1/\nu$, but prices, transfers, tenure, individual fertility, housing, and population scale can differ across steady states.*

Existence follows under the boundary and continuity conditions in Appendix A.3; local uniqueness requires the two-by-two Jacobian of (15) to be nonsingular. The steady state is thus an equilibrium implication, not a normalization. Replacement fixes average fertility only after imposing stationarity.

Suppose ϑ_t rises permanently at date zero and the economy starts at the old positive steady state. The predetermined state is $S_t = (Y_t, O_t, G_t, P_{t-1})$, where G_t is the distribution of old financial wealth, inherited housing, and tenure. Given a candidate perfect-foresight sequence, household policies generate the two date- t residuals

$$\mathcal{R}_t^H = Y_t \bar{h}_t^Y + O_t \bar{h}_t^O - \bar{H}, \quad (16)$$

$$\mathcal{R}_t^G = (Y_t + O_t)T_t - \tau^p P_t \bar{H} - \ell_t O_t \bar{s}_t^O. \quad (17)$$

A transition equilibrium is a root of this infinite sequence together with the cohort and distribution laws, converging to the new steady state. Computation uses terminally closed finite-horizon truncations: prices, transfers, and value functions are fixed at the new steady state beyond the terminal date, while the terminal endogenous state is allowed to approach it. A regime-consistent price branch and a nonsingular stacked Jacobian give a locally unique truncation for sufficiently small permanent shocks; explicit boundary signs give global existence of a truncation. A convergent sequence of such roots is an exact infinite-horizon transition under the uniform-tail condition in Appendix A.4.

The capital-gains basis needs no second shock. If equilibrium prices satisfy $P_s > P_{s-1}$ on several successive dates, then $\ell_s = \tau^g(P_s - P_{s-1}) > 0$ for each corresponding old cohort. A two-period life makes the preceding date's price the basis; it does not restrict the number of dates over which the equilibrium price adjusts.

1.4 Intergenerational reallocation and fertility

Consider an old incumbent at an interior retention choice and a young owner whose down-payment constraint binds while the owner-size limit is slack. Their marginal values of one housing unit, in current goods, are

$$MV_{it}^Y = r_t + q\ell_{t+1} + \zeta_{it}^{O,F}, \quad MV_{jt}^O = r_t - \ell_t. \quad (18)$$

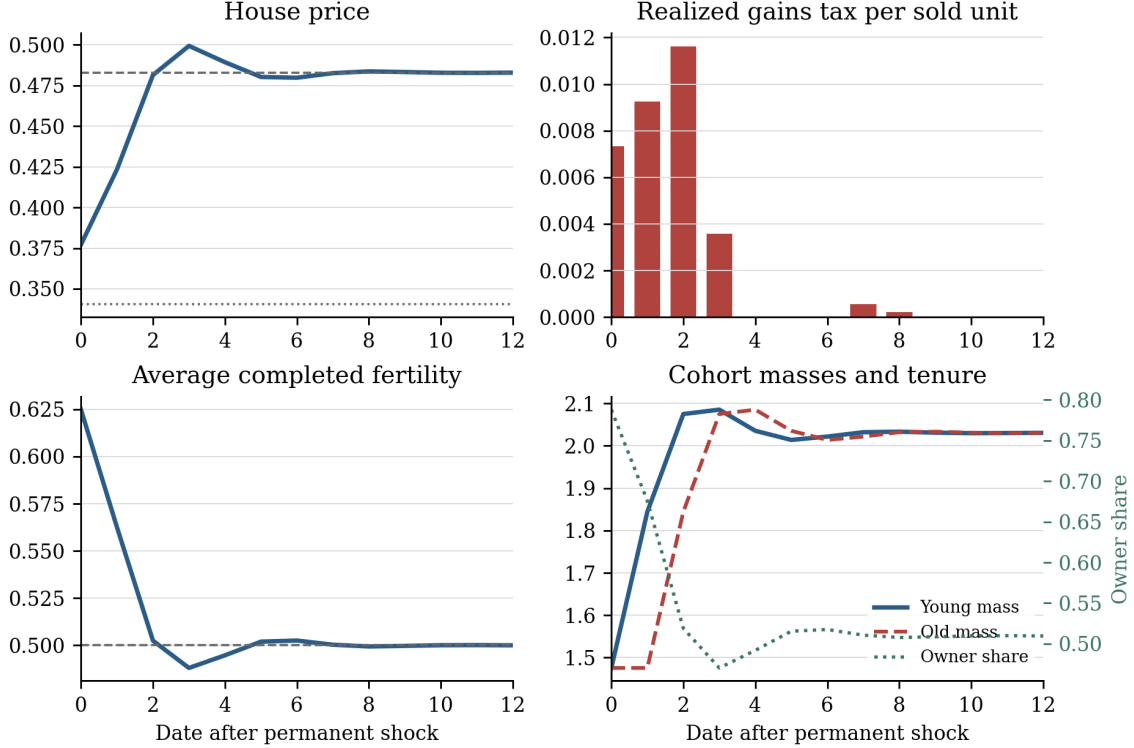


Figure 1: A terminally closed mixed-tenure transition approximation after one permanent increase in the fertility preference. Dotted and dashed lines in the price panel are the old and new steady-state prices; the horizontal fertility line is replacement $1/\nu$. The model has positive renter and owner masses on every displayed date. The preference change is the only exogenous shock.

Proposition 2 (Conditional intergenerational reallocation). *Fix fertility, cohort masses, tenure, and all unaffected household allocations. Suppose a planner can transfer the marginal unit's title and occupancy, provide pledgeable bridge finance, compensate the incumbent with lump-sum transfers, and reclaim the incremental gains-tax rebates at dates t and $t+1$. If the real transfer cost is K_{ijt} , moving a sufficiently small unit from old incumbent j to constrained young buyer i is a compensated improvement when*

$$\underbrace{\zeta_{it}^{O,F} + \ell_t + q\ell_{t+1}}_{MV_{it}^Y - MV_{jt}^O} > K_{ijt}. \quad (19)$$

At a steady state the condition reduces to $\zeta_i^{O,F} > K_{ij}$.

This is deliberately a local allocation result. A reform that changes fertility changes the number of future households, so a global ranking also requires an explicit social criterion for comparing populations. The reallocation proposition instead fixes population and identifies the exact intergenerational wedge before choosing such a criterion.

Housing access nevertheless has a precise fertility implication. Holding tenure, prices, lifetime wealth, and the active set fixed, let $D = \mathcal{W}_{it} - r_t^m \hat{h} - \chi n$ and $S = \hat{h} - \kappa n$. Differentiating (11) gives

$$\frac{dn}{d\hat{h}} = \frac{\alpha\kappa/S^2 - (1+k)\chi r_t^m/D^2}{\vartheta_t/n^2 + (1+k)\chi^2/D^2 + \alpha\kappa^2/S^2}. \quad (20)$$

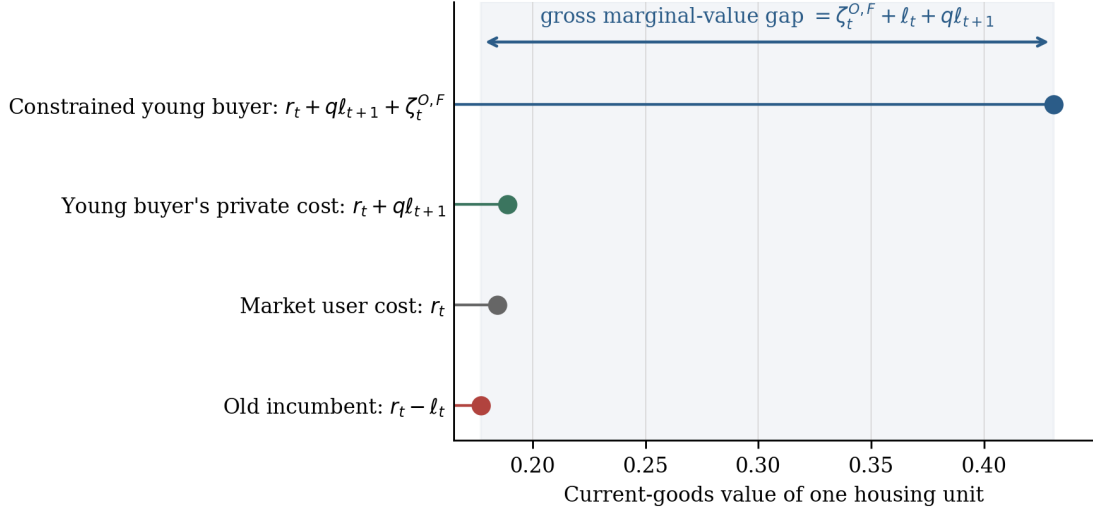


Figure 2: Marginal values on the computed path. The financing wedge raises the young buyer's value; realization taxes lower the old incumbent's value and raise the buyer's anticipated private cost. The displayed object is the gross marginal-value gap. Under the ledger in Proposition 2, a compensated improvement obtains when this gap exceeds the real transfer cost.

The denominator is positive. Using the housing first-order condition, the exact sign restriction is

$$\frac{dn}{d\hat{h}} \geq 0 \iff \chi \leq \frac{(1+k)\kappa(r_t^m + \zeta_{it}^m)^2}{\alpha r_t^m}. \quad (21)$$

More accessible space raises fertility when children are sufficiently space-intensive relative to their goods cost. This partial-equilibrium sign does not by itself sign a funded policy: prices, rebates, tenure, and lifetime resources also respond in general equilibrium.

A Proofs and computational construction

This appendix derives the conditional household policies, gives sufficient conditions for positive steady states and finite-horizon transition truncations, states conditions under which their limit is an exact transition equilibrium, proves the reallocation and fertility results, and reports a reproducible mixed-tenure transition approximation.

A.1 Old-age problems

An old renter with financial wealth a solves

$$\begin{aligned} V_t^R(a) = \max_{c^O, h^O, e} \quad & \log c^O + \gamma \log h^O + \omega_B \log e \\ \text{s.t.} \quad & c^O + qe + r_t h^O = a + T_t, \quad 0 < h^O \leq \bar{h}^R. \end{aligned} \quad (22)$$

For an old owner, define

$$A = 1 + \gamma + \omega_B, \quad \varrho_t = r_t - \ell_t, \quad X_t = a + (P_t - \ell_t)H + T_t. \quad (23)$$

The owner problem is (5). Both old problems are strictly concave on their feasible sets.

Lemma 1 (Old-age allocations). *Suppose $r_t > 0$, $\varrho_t > 0$, and resources place the household in the log domain. If the renter size limit is slack, then*

$$c^O = \frac{a + T_t}{A}, \quad h^O = \frac{\gamma(a + T_t)}{A r_t}, \quad e = \frac{\omega_B(a + T_t)}{A q}. \quad (24)$$

If the renter limit binds, set $Z^R = a + T_t - r_t \bar{h}^R$; then

$$h^O = \bar{h}^R, \quad c^O = \frac{Z^R}{1 + \omega_B}, \quad e = \frac{\omega_B Z^R}{(1 + \omega_B)q}. \quad (25)$$

If both owner constraints are slack, then

$$c^O = \frac{X_t}{A}, \quad h^O = \frac{\gamma X_t}{A \varrho_t}, \quad e = \frac{\omega_B X_t}{A q}, \quad (26)$$

and the two strict slackness conditions are

$$\frac{\gamma X_t}{A \varrho_t} < H, \quad \omega_B \varrho_t > \gamma q P_{t+1}. \quad (27)$$

Proof. For a generic interior budget $c^O + qe + p_h h^O = X$, the three first-order conditions imply $p_h h^O = \gamma c^O$ and $qe = \omega_B c^O$. The budget gives $A c^O = X$. Substituting $p_h = r_t$ for renters and $p_h = \varrho_t$ for owners gives (24) and (26). If renter housing is fixed at $\bar{h}^R$, strict concavity allocates the remaining resources between consumption and the estate in shares $1 : \omega_B$, which gives (25). Substitution of the interior owner policy into $h^O < H$ and $e > P_{t+1} h^O$ yields (27). $\square$

The remaining owner active sets are also explicit. If only estate composition binds, let $d_t = \varrho_t + q P_{t+1}$; then

$$c^O = \frac{X_t}{A}, \quad h^O = \frac{(\gamma + \omega_B) X_t}{A d_t}, \quad e = P_{t+1} h^O. \quad (28)$$

If only retention binds, set $Z_t = X_t - \varrho_t H$; then

$$h^O = H, \quad c^O = \frac{Z_t}{1 + \omega_B}, \quad e = \frac{\omega_B Z_t}{(1 + \omega_B)q}. \quad (29)$$

If both constraints bind,

$$h^O = H, \quad e = P_{t+1}H, \quad c^O = X_t - (\varrho_t + qP_{t+1})H. \quad (30)$$

The estate-only row requires $\gamma q P_{t+1} \geq \omega_B \varrho_t$ and $h^O < H$. The retention-only row requires

$$\omega_B Z_t > (1 + \omega_B)qP_{t+1}H, \quad \gamma Z_t \geq (1 + \omega_B)\varrho_t H. \quad (31)$$

For the both-binding row, the multipliers are nonnegative exactly when

$$\frac{q}{c^O} \geq \frac{\omega_B}{P_{t+1}H}, \quad \frac{\gamma + \omega_B}{H} \geq \frac{\varrho_t + qP_{t+1}}{c^O}. \quad (32)$$

These rows and complementary slackness exhaust the strictly concave owner problem.

In the interior owner region, the indirect value and relevant envelopes are

$$V_t^O(a, H) = A \log X_t - A \log A + \gamma \log \gamma - \gamma \log \varrho_t + \omega_B \log \omega_B - \omega_B \log q, \quad (33)$$

$$V_{a,t}^O = \frac{A}{X_t} = \frac{1}{c_t^O}, \quad V_{H,t}^O = (P_t - \ell_t)V_{a,t}^O. \quad (34)$$

A.2 Young household solutions and tenure

The renter and owner problems are those in the main text. Let λ_t^m be the current-budget multiplier, η_t^m the multiplier on the tenure-specific housing limit, v_t^m the multiplier on $a' \geq 0$, and μ_t the owner down-payment multiplier. The common consumption and fertility conditions are

$$\lambda_t^m = \frac{1}{c_t^m}, \quad (35)$$

$$\frac{\vartheta_t}{n_t^m} = \lambda_t^m \chi + \frac{\alpha \kappa}{h_t^m - \kappa n_t^m}. \quad (36)$$

The saving conditions are

$$\lambda_t^R = \beta R V_{a,t+1}^R + v_t^R, \quad (37)$$

$$\lambda_t^O = \beta R V_{a,t+1}^O + v_t^O, \quad v_t^m a_t^m = 0. \quad (38)$$

When saving and next-period old choices are interior, the housing conditions reduce to

$$\frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_t^m, \quad \zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \quad \zeta_t^O = \frac{\mu_t}{\lambda_t^O} (1 - \phi) P_t + \frac{\eta_t^O}{\lambda_t^O}. \quad (39)$$

This proves (12). If the owner-size limit is slack, the second term in ζ_t^O is zero and the access wedge is purely financial.

Proposition 3 (Lifetime reduction and conditional policies). *Suppose young saving and next-period old constraints are interior. Then the conditional lifetime budget is (8). If the housing limit is also slack, the policies are (9) and saving is*

$$a_t'^R = kc_t^R - qT_{t+1}, \quad (40)$$

$$a_t'^O = kc_t^O - qT_{t+1} - [q(P_{t+1} - \ell_{t+1}) - \phi P_t]h_t^O. \quad (41)$$

If the effective housing limit binds, $h_t^m = \hat{h}_{it}^m$ and fertility is the unique root of (11) in

$$0 < n < \min \left\{ \frac{\hat{h}_{it}^m}{\kappa}, \frac{\mathcal{W}_{it} - r_t^m \hat{h}_{it}^m}{\chi} \right\}. \quad (42)$$

Proof. The saving Euler equation and Lemma 1 imply $c_{t+1}^O = \beta R c_t$ and therefore $qX_{t+1} = kc_t$. For an owner,

$$X_{t+1} = Ra_t' - R\phi P_t h_t + (P_{t+1} - \ell_{t+1})h_t + T_{t+1}. \quad (43)$$

Substitute this expression and the current owner budget into $qX_{t+1} = kc_t$. The mortgage inflow and repayment cancel, leaving $(1+k)c_t + \chi n_t + [r_t + q\ell_{t+1}]h_t = \mathcal{W}_{it}$. The renter derivation gives the same identity with $r_t^R = r_t$.

Let $s = h - \kappa n$. When housing is unconstrained, (39) gives $s = \alpha c / r_t^m$ and (36) gives $n = \vartheta_t c / (\chi + \kappa r_t^m)$. Substitution into the lifetime budget gives $\mathcal{W}_{it} = (1+k+\alpha+\vartheta_t)c$, proving (9). The identity $qX_{t+1} = kc_t$ gives (40)–(41).

When the limit binds, substitute $h = \hat{h}$ and $c = (\mathcal{W} - r_t^m \hat{h} - \chi n) / (1+k)$ into the fertility condition. This gives (11). Its left side tends to $+\infty$ at the lower boundary of (42), tends to $-\infty$ at the upper boundary, and has derivative

$$-\frac{\vartheta_t}{n^2} - \frac{\alpha \kappa^2}{(\hat{h} - \kappa n)^2} - \frac{(1+k)\chi^2}{(\mathcal{W} - r_t^m \hat{h} - \chi n)^2} < 0. \quad (44)$$

The root therefore exists and is unique. $\square$

If saving is zero or a different old-age active set applies, the lifetime reduction is replaced by the KKT system, but strict concavity still makes each conditional tenure policy unique. The paper results state their active-set conditions explicitly rather than extending closed forms through those corners.

With the type-I extreme-value shocks in (6), the conditional probability of ownership is strictly between zero and one whenever the two deterministic values are finite. Because the logit formula and the conditional policies are continuous, aggregate choices are continuous across the deterministic rent-own indifference locus. An atomless continuous tenure taste distribution gives the same regularity without the logit specialization.

A.3 Aggregation and positive steady states

Let F denote the distribution of observed young types $i = (y_i, b_i)$. The old-state distribution is over financial wealth a , inherited owner housing H , and tenure. For any bounded measurable test function ψ , its law of motion is

$$\int \psi dG_{t+1} = \int \left[(1 - \pi_t^O(i)) \psi(Ra_t'^R(i), 0, R) \right]$$

$$+ \pi_t^O(i) \psi(Ra_t^O(i) - R\phi P_t h_t^O(i), h_t^O(i), O) \Big] dF(i). \quad (45)$$

Average fertility and young housing integrate the corresponding conditional policies with the same weights. Average old housing integrates the old policies over G_t . Average taxable sales are

$$\bar{s}_t^O = \int \mathbf{1}\{m = O\} [H - h_t^O(a, H, O)] dG_t. \quad (46)$$

Equations (13)–(14), the cohort laws, and (45) close the economy.

Proof of Proposition 1. In a steady state, $O_{t+1} = Y_t$ implies $O^* = Y^*$. Stationarity of the young cohort then implies $1 = \nu\mathcal{N}(P^*, T^*; \vartheta)$. A constant real price makes $\ell^* = 0$ directly from (3). Housing clearing gives $Y^* = \bar{H}/\mathcal{H}$. The government budget becomes $2Y^*T^* = \tau^p P^* \bar{H}$. Eliminating Y^* gives the second equation in (15). Reversing these steps establishes sufficiency once the household policies and stationary distribution are constructed. $\square$

The following result gives checkable sufficient conditions for existence. It also makes clear that a positive steady state is not guaranteed for arbitrary primitives.

Proposition 4 (Steady-state existence and local uniqueness). *Let $\mathcal{D} = [\underline{P}, \bar{P}] \times [\underline{T}, \bar{T}] \subset \mathbb{R}_{++}^2$ be a rectangle on which conditional policies are feasible, relevant user costs are positive, and $\mathcal{N}$ and $\mathcal{H}$ are continuous. Define*

$$\Psi(P, T) = \frac{\tau^p P}{2} \mathcal{H}(P, T; \vartheta). \quad (47)$$

Suppose:

1. $\Psi(P, \cdot)$ maps $[\underline{T}, \bar{T}]$ into itself for every P , and $\sup_{\mathcal{D}} |\Psi_T| < 1$;
2. writing $T = \mathcal{T}(P)$ for its unique fixed point,

$$[\nu\mathcal{N}(\underline{P}, \mathcal{T}(\underline{P}); \vartheta) - 1][\nu\mathcal{N}(\bar{P}, \mathcal{T}(\bar{P}); \vartheta) - 1] < 0. \quad (48)$$

Then a positive steady state exists in $\mathcal{D}$. It is locally unique if

$$J^* = \begin{bmatrix} \nu\mathcal{N}_P & \nu\mathcal{N}_T \\ -\frac{\tau^p}{2}(\mathcal{H} + P\mathcal{H}_P) & 1 - \frac{\tau^p P}{2}\mathcal{H}_T \end{bmatrix}_{(P^*, T^*; \vartheta)} \quad (49)$$

is nonsingular.

Proof. The contraction theorem gives a unique fiscal fixed point $\mathcal{T}(P)$ for each P . Uniform contraction and continuity of Ψ imply continuity of $\mathcal{T}$. The intermediate value theorem applied to the replacement residual along this locus gives a price P^* satisfying the first steady-state equation. Its fiscal fixed point $T^* = \mathcal{T}(P^*)$ satisfies the second. Local uniqueness follows from the implicit-function theorem when J^* is nonsingular. $\square$

The first condition can be checked from bounds on primitives and policies. For example, if $0 < \underline{\mathcal{H}} \leq \mathcal{H} \leq \bar{\mathcal{H}}$ on $\mathcal{D}$, it is enough to choose transfer bounds such that

$$\underline{T} \leq \frac{\tau^p \underline{P}}{2} \underline{\mathcal{H}}, \quad \bar{T} \geq \frac{\tau^p \bar{P}}{2} \bar{\mathcal{H}}, \quad \frac{\tau^p \bar{P}}{2} \sup_{\mathcal{D}} |\mathcal{H}_T| < 1. \quad (50)$$

The replacement boundary condition says that the low- and high-price faces place average fertility on opposite sides of $1/\nu$ after closing the fiscal account. It is an inequality in the unique household policies, and hence is directly verifiable from primitives rather than an assumed equilibrium.

If J^* is nonsingular, the response to a permanent change in ϑ is

$$\begin{bmatrix} P_\vartheta^* \\ T_\vartheta^* \end{bmatrix} = -(J^*)^{-1} \begin{bmatrix} \nu \mathcal{N}_\vartheta \\ -\frac{\tau^p P^*}{2} \mathcal{H}_\vartheta \end{bmatrix}. \quad (51)$$

The associated population-scale response is

$$\frac{d \log Y^*}{d\vartheta} = -\frac{\mathcal{H}_P P_\vartheta^* + \mathcal{H}_T T_\vartheta^* + \mathcal{H}_\vartheta}{\mathcal{H}}. \quad (52)$$

There is no general sign without restrictions on the demand and tenure elasticities.

A.4 Transition equilibrium

Suppose $\vartheta_t = \vartheta^0$ for $t < 0$ and $\vartheta_t = \vartheta^1$ for $t \geq 0$, and both values support regular positive steady states $\mathcal{E}^0$ and $\mathcal{E}^1$. The date-zero state is the old steady state: $S_0 = (Y^0, Y^0, G^0, P^0)$. In particular, $P_{-1} = P^0$ is the basis of date-zero old owners. No additional initialization shock is present.

The gains rule is equivalently the complementarity system

$$\ell_t \geq 0, \quad \ell_t \geq \tau^g(P_t - P_{t-1}), \quad \ell_t[\ell_t - \tau^g(P_t - P_{t-1})] = 0. \quad (53)$$

It is continuous but kinked. Within an appreciating regime its price derivative is τ^g ; within a non-appreciating regime it is zero.

Fix a terminal date J and use the new steady-state prices, transfers, and continuation values from $J+1$ onward. The state generated at $J+1$ is not separately forced to equal the new steady-state state. The resulting object is therefore a terminally closed truncation, not necessarily an exact finite prefix of an infinite-horizon transition. Let z_J stack $(\log P_0, \log T_0, \dots, \log P_J, \log T_J)$, with residuals interleaved in the same order. Recursively applying household policies, the distribution law, and cohort laws turns (17) into a finite system

$$\mathbf{R}_J(z_J; \varepsilon) = 0, \quad (54)$$

where ε scales the permanent change from ϑ^0 to ϑ^1 and the terminal steady state also varies with ε .

Because the constant path lies at the gains kink, local analysis is directional. Choose a candidate appreciation pattern $d = (d_0, \dots, d_{J+1}) \in \{0, 1\}^{J+2}$ and define the smooth branch

$$\ell_t^d = \tau^g d_t (P_t - P_{t-1}). \quad (55)$$

It is a branch of the statutory rule when $P_t - P_{t-1} > 0$ for $d_t = 1$ and $P_t - P_{t-1} < 0$ for $d_t = 0$. Let $\mathbf{R}_J^d$ denote the residual system evaluated on this branch.

Proposition 5 (Directional local existence of a truncation). *Suppose the old steady state is regular, the household active sets are strict, and the branch Jacobian $\partial \mathbf{R}_J^d / \partial z_J$ is nonsingular at $\varepsilon = 0$. Suppose also that the directional solution has*

$$\left. \frac{d(P_t - P_{t-1})}{d\varepsilon} \right|_0 > 0 \text{ if } d_t = 1, \quad \left. \frac{d(P_t - P_{t-1})}{d\varepsilon} \right|_0 < 0 \text{ if } d_t = 0. \quad (56)$$

Then, for each fixed J , a unique branch-consistent terminally closed truncation exists for all sufficiently small positive shocks, varies continuously differentiably with the shock, and satisfies the statutory gains rule on dates $0, \dots, J$.

Proof. Steady-state regularity and Proposition 4 give a continuously differentiable terminal steady state for small ε . Strict active sets, positive log domains, the logit tenure share, and (55) make $\mathbf{R}_J^d$ continuously differentiable in (z_J, ε) . At $\varepsilon = 0$, the constant old-steady-state path is a root on every branch. Nonsingularity and the implicit-function theorem give a unique differentiable branch solution. The strict directional signs in (56) persist for sufficiently small positive ε and validate every selected branch. Therefore $\ell_t^d = \tau^g[P_t - P_{t-1}]_+$ and the branch solution solves the truncated statutory system. $\square$

If a directional price difference is zero, ordinary branch analysis is inconclusive at that date; one must use higher-order terms, a generalized Jacobian, or the continuous global existence result below. Thus the local proposition does not hide the steady-state kink inside a differentiability assumption.

A sufficient derivative condition for nonsingularity is strict row diagonal dominance,

$$\left| \frac{\partial R_{J,k}}{\partial z_{J,k}} \right| > \sum_{\ell \neq k} \left| \frac{\partial R_{J,k}}{\partial z_{J,\ell}} \right| \quad \text{for every } k. \quad (57)$$

All terms are derivatives of the explicit household policies and state laws. Diagonal dominance is sufficient, not necessary; the numerical construction below checks the singular values of the full stacked Jacobian directly.

The next result provides a global existence criterion for the truncated system that does not presuppose a clearing self-map.

Proposition 6 (Boundary-sign existence of a truncation). *Let $\mathcal{B}_J = \prod_{k=1}^{2(J+1)} [z_k, \bar{z}_k]$ be a box on which resources and user costs are positive, household policies are well-defined, and the induced states remain in a compact set. Suppose the stacked residual is continuous on $\mathcal{B}_J$ and, for every coordinate k , either*

$$R_{J,k}(z_k, z_{-k}) \leq 0 \leq R_{J,k}(\bar{z}_k, z_{-k}) \quad \text{for every } z_{-k}, \quad (58)$$

or the two inequalities are reversed. Then a terminally closed truncation exists in $\mathcal{B}_J$.

Proof. The Poincaré–Miranda theorem applied to the continuous vector $\mathbf{R}_J$ gives a zero in the box. The max operator in (53) preserves continuity, so the argument allows appreciation kinks even though ordinary derivatives fail there. $\square$

The boundary signs have an economic interpretation. The price faces require housing excess demand and excess supply at the respective bounds, after all forward-looking choices and inherited states are evaluated. The transfer faces require revenues to exceed rebates at the lower bound and rebates to exceed revenues at the upper bound. These inequalities are sufficient and must be verified for the parameterization; the model does not guarantee them for arbitrary primitives.

An infinite-horizon result requires an additional limiting argument.

Proposition 7 (Limit of terminally closed truncations). *Let $x_t^J = (P_t^J, T_t^J, S_t^J)$ denote a root of the horizon- J truncated system. Suppose these roots lie in a common compact set, their induced state distributions are tight, and there is a sequence $J_k \rightarrow \infty$ along which $x_t^{J_k}$ converges for every fixed*

date t (with weak convergence for the distribution component). Suppose also that the roots satisfy the uniform-tail condition

$$\lim_{L \rightarrow \infty} \sup_{J \geq L} \sup_{L \leq t \leq J} d(x_t^J, \mathcal{E}^1) = 0. \quad (59)$$

If household policies and state laws are continuous in these topologies, the pointwise limit is an infinite-horizon transition equilibrium converging to $\mathcal{E}^1$.

Proof. Fix any date t . For all sufficiently large k , the horizon- J_k root satisfies the date- t household, market-clearing, government, cohort, and distribution equations. Passing to the limit preserves these equations by continuity. Because t was arbitrary, the pointwise limit satisfies every finite-date equilibrium condition. Equation (59) implies convergence of that limit to $\mathcal{E}^1$. $\square$

The horizon and terminal checks in the numerical construction are evidence compatible with this convergence argument, but two horizons cannot verify the uniform-tail condition and hence do not prove an infinite-horizon equilibrium.

Corollary 1 (One permanent shock and multiple taxable gains). *If the equilibrium following the single permanent shock has $P_s > P_{s-1}$ for $s = t, \dots, t + L$, then $\ell_s > 0$ for all $L + 1$ dates.*

Proof. Equation (3) gives $\ell_s = \tau^g(P_s - P_{s-1}) > 0$ on each appreciating date. The two-period life determines the one-period basis P_{s-1} of cohort s ; it places no restriction on the length of the aggregate price transition. $\square$

A.5 Conditional reallocation

For a constrained young owner, (39) gives the current-goods marginal value

$$MV_{it}^Y = \frac{\alpha c_{it}^O}{h_{it}^O - \kappa n_{it}^O} = r_t + q\ell_{t+1} + \zeta_{it}^{O,F}. \quad (60)$$

For an old incumbent at an interior retention choice, the housing first-order condition from (5) gives

$$MV_{jt}^O = \frac{\gamma c_{jt}^O}{h_{jt}^O} = r_t - \ell_t. \quad (61)$$

Their difference is the left side of (19) before transaction costs.

Assumption 1 (Two-date transfer ledger). For an infinitesimal transfer of housing from incumbent j to buyer i , the allocation authority can: (i) transfer both title and occupancy at date t ; (ii) provide pledgeable bridge finance without relaxing other households' constraints; (iii) make household-specific lump-sum transfers at t and $t + 1$; (iv) reclaim the incremental equal gains-tax rebates on the transferred unit at both dates; and (v) ensure the buyer's next-period retention constraint is slack, so the marginal unit is resold at $t + 1$. Fertility, tenure, cohort masses, and all unaffected allocations are held fixed.

Proof of Proposition 2. Transfer $\epsilon > 0$ units from the old incumbent to the young buyer. At the baseline allocation, the young buyer is willing to surrender $MV_{it}^Y \epsilon + o(\epsilon)$ current goods without losing utility, while the old incumbent requires $MV_{jt}^O \epsilon + o(\epsilon)$ current goods to be compensated. The bridge makes the transfer feasible despite the down-payment constraint. The old sale realizes

$\ell_t \epsilon$ of date- t revenue; the buyer's marginal resale realizes $\ell_{t+1} \epsilon$ at $t + 1$. Under Assumption 1, the authority reclaims the associated rebates, whose date- t value is $\ell_t \epsilon + q \ell_{t+1} \epsilon$.

After paying the real resource cost $K_{ijt} \epsilon$, the first-order amount available for compensation is

$$[MV_{it}^Y - MV_{jt}^O - K_{ijt}] \epsilon = [\zeta_{it}^{O,F} + \ell_t + q \ell_{t+1} - K_{ijt}] \epsilon. \quad (62)$$

If the bracket is strictly positive, transfers can leave both affected households weakly above baseline and one strictly above it. Smoothness and the strict first-order inequality imply the same result for a sufficiently small finite ϵ . At a steady state, $\ell_t = \ell_{t+1} = 0$, giving the stated condition $\zeta_i^{O,F} > K_{ij}$. $\square$

The transfer ledger is part of the proposition, not an institutional detail. Without pledgeable bridge finance or two-date transfers, the marginal-value gap remains a diagnostic of misallocation but need not be implementable as a Pareto improvement. The proposition also fixes population. Policies that change fertility compare economies with different sets of people, for which a social criterion must be stated explicitly; see Golosov et al. (2007) and Pérez-Nievas et al. (2019). The paper therefore reports the conditional allocation result rather than a global welfare theorem.

A.6 Housing access and fertility

Suppose $h = \hat{h}$ binds and fertility remains interior. Define

$$D = \mathcal{W}_{it} - r_t^m \hat{h} - \chi n, \quad S = \hat{h} - \kappa n, \quad c = \frac{D}{1+k}. \quad (63)$$

Derivation of (20) and (21). Let the left side of (11) be $F(n, \hat{h})$. Direct differentiation gives

$$F_n = -\frac{\vartheta_t}{n^2} - \frac{\alpha \kappa^2}{S^2} - \frac{(1+k)\chi^2}{D^2} < 0, \quad (64)$$

$$F_{\hat{h}} = \frac{\alpha \kappa}{S^2} - \frac{(1+k)\chi r_t^m}{D^2}. \quad (65)$$

The implicit-function theorem gives $dn/d\hat{h} = -F_{\hat{h}}/F_n$, which is (20). At a constrained allocation, (39) and $c = D/(1+k)$ imply

$$\frac{D}{S} = \frac{(1+k)(r_t^m + \zeta_{it}^m)}{\alpha}. \quad (66)$$

Substitute this ratio into $F_{\hat{h}} \geq 0$ and rearrange to obtain (21). $\square$

A primitive sufficient condition, stronger than the exact condition, is

$$\chi \leq \frac{(1+k)\kappa r_t^m}{\alpha}, \quad (67)$$

because $\zeta_{it}^m \geq 0$. When the down-payment constraint alone binds,

$$\hat{h}_{it}^O = \frac{b_i}{(1-\phi)P_t}, \quad \frac{\partial \hat{h}^O}{\partial b_i} = \frac{1}{(1-\phi)P_t}, \quad \frac{\partial \hat{h}^O}{\partial \phi} = \frac{b_i}{(1-\phi)^2 P_t}. \quad (68)$$

These expressions give the direct fertility response to a local relaxation. A funded policy also moves prices, transfers, lifetime wealth, and tenure, so its aggregate response requires the equilibrium system.

A.7 Solved mixed-tenure transition approximation

The numerical construction specializes observed types to common income and liquid wealth while retaining the iid tenure tastes in (6). It gives exact steady-state equilibria and a terminally closed transition approximation; it is neither a calibration nor a numerical proof of the infinite-horizon limiting conditions. Table 1 lists the parameters.

Table 1: Parameters of the mixed-tenure construction

Object	Parameter	Value	Object	Parameter	Value
Bond price	q	0.50	Mortgage share	ϕ	0.80
Property tax	τ^P	0.05	Gains-tax rate	τ^g	0.20
Space weight	α	0.40	Space per child	κ	0.50
Goods per child	χ	0.15	Young discount	β	0.40
Old housing weight	γ	0.30	Estate weight	ω_B	0.40
Income	y	1.00	Liquid wealth	b	0.06
Household formation	ν	2.00	Housing stock	$\bar{H}$	2.00
Rental limit	$\bar{h}^R$	0.45	Owner limit	$\bar{h}^O$	2.00
Owner intercept	$\bar{\xi}$	-0.15	Taste scale	κ_T	0.12

At date zero, ϑ rises permanently from 0.35 to 0.50. Solving (15) gives Table 2.

Table 2: Positive steady-state endpoints

Object	Initial steady state	Final steady state
Fertility preference ϑ	0.3500	0.5000
Average fertility $\bar{n}$	0.5000	0.5000
House price P	0.3405	0.4829
Per-household transfer T	0.01154	0.01189
Young owner share	0.7419	0.5096
Renter fertility	0.3511	0.4366
Owner fertility	0.5518	0.5610
Young renter housing	0.4500	0.4500
Young owner housing	0.8810	0.6212
Old renter housing	0.4500	0.4500
Old owner housing	0.6327	0.4457
Young and old cohort mass	1.4757	2.0304

Average fertility is $1/\nu = 0.5$ at both endpoints, but conditional fertility, tenure, housing, prices, and population scale differ. The determinants of the two steady-state Jacobians, computed in $(\log P, \log T)$, are -0.00913 and -0.00838 , so both roots are locally regular.

The horizon-28 truncation uses the final steady-state continuation values, prices, and transfers from date 29 onward and solves all date-by-date housing and government equations through date 28 jointly. The state generated at date 29 is recorded rather than imposed. On impact, average fertility rises to 0.6248; the young cohort then expands with the OLG lag. The price moves from 0.3405 before the shock to 0.3772 on impact, continues to 0.4994 by date 3, and then approaches 0.4829 with damped oscillations. Realized gains taxes are materially positive on each of dates 0 through 3, even though the economy receives only the date-zero permanent preference shock.

The computed truncation has both tenures on every displayed date: the owner share lies between

0.4704 and 0.7877. Minimum renter and owner saving are 0.3530 and 0.4039. The unconstrained young renter and owner housing policies exceed their respective limits by at least 0.5081 and 0.3574, so both young limits bind strictly. The owner-size limit has slack of at least 1.2046, so the owner wedge is purely financial. The old renter cap has a minimum shadow wedge of 0.00676. Old owners sell at least 0.1663 units, their incumbent user cost is at least 0.1770, and their estate-composition slack is at least 0.0228; hence the old-owner interior formulas apply strictly throughout.

The maximum absolute scaled housing or government residual is 2.07×10^{-10} . The maximum terminal-state discrepancy is 1.10×10^{-8} . Replacing horizon 28 by horizon 24 changes the first nine dates in prices, transfers, masses, tenure, fertility, and housing by at most 1.84×10^{-15} . At the computed horizon-28 root, freezing the realized appreciation pattern gives a smallest stacked-Jacobian singular value of 0.401 and a condition number of 48.6. These are conditioning diagnostics for that finite root, not a verification of the directional derivative assumptions at a zero-size shock. Finally, the analytical fertility derivative is 0.6490218128; a centered finite difference gives 0.6490218128, with absolute error 8.0×10^{-12} .

The construction is generated by `code/model/tools/build_simplified_olg_mixed_tenure_theory.py`. It writes the steady states, full truncated path, verification ledger, and the two figures used in the paper. These files make the finite-root and active-set claims reproducible; they do not replace the limiting conditions in Proposition 7.

References

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